

**Healthcare Consulting Engagement Packet
National Hospital Network (NHN)**

1. CLIENT ENGAGEMENT LETTER

National Hospital Network
Data Science Consulting Engagement

CONFIDENTIAL CLIENT ENGAGEMENT

Date: July 2026

To: Data Science Consulting Team

From: Dr. Sarah Mitchell, Chief Medical Officer

Executive Summary

National Hospital Network (NHN) is one of the largest integrated healthcare systems in the United States, operating 42 hospitals, more than 300 outpatient facilities, and serving millions of patients annually across 11 states.

Over the past four years, NHN has experienced a significant increase in 30-day hospital readmission rates, creating growing concern among Executive Leadership. While some readmissions are unavoidable, the current trend has resulted in measurable operational and financial challenges, including:

- Increased operating costs
- Reduced patient satisfaction scores
- Increased strain on physicians, nurses, and clinical staff
- Higher utilization of inpatient resources
- Increased pressure on care coordination teams
- Potential financial penalties under Centers for Medicare & Medicaid Services (CMS) reimbursement programs

Executive Leadership believes that these challenges cannot be explained by a single factor.

Instead, they believe rising readmissions are likely influenced by a combination of clinical, operational, financial, and post discharge care coordination issues.

Although NHN has invested significantly in quality improvement initiatives, care coordination programs, discharge planning, and patient follow-up services, readmission rates continue to increase across multiple patient populations.

To address these challenges, the Executive Leadership Team has commissioned this consulting engagement to identify the primary drivers of hospital readmissions and develop a comprehensive, data driven strategy that improves patient outcomes while reducing avoidable healthcare costs.

Your consulting team has been selected to conduct this engagement. Throughout this engagement, you will analyze patient level clinical information, evaluate financial impacts associated with readmissions, investigate the effectiveness of current care coordination programs, develop predictive machine learning models, build executive dashboards, and provide strategic recommendations designed to improve patient outcomes and strengthen organizational performance.

The results of this engagement will be presented directly to the Executive Leadership Team and the National Hospital Network Board of Directors. Leadership intends to use your recommendations to guide future investments in patient care, discharge planning, care coordination, operational improvement initiatives, and data driven decision making.

Executive Sponsor

Dr. Sarah Mitchell
Chief Medical Officer
National Hospital Network

Engagement Objectives

Your consulting team has been retained to:

1. Analyze patient, operational, and financial data related to hospital readmissions.
2. Identify the primary clinical, operational, and organizational drivers contributing to increased readmission rates.
3. Develop predictive machine learning models capable of identifying patients at high risk of readmission before discharge.
4. Evaluate the effectiveness of current care coordination and post discharge intervention programs.
5. Quantify the financial impact of avoidable readmissions, including estimated costs, reimbursement implications, and potential return on investment (ROI) associated with proposed interventions.
6. Develop interactive executive dashboards that provide leadership with actionable insights into patient risk, operational performance, financial impact, and care coordination effectiveness.
7. Present strategic recommendations that improve patient outcomes, reduce avoidable readmissions, strengthen care coordination, and support National Hospital Network's long-term quality and financial objectives.

Expected Deliverables

Technical Deliverables

- Exploratory Data Analysis (EDA)
- Hospital Readmission Prediction Model
- Patient Risk Stratification Analysis
- Care Coordination Effectiveness Analysis
- Executive Dashboard (Tableau or Power BI)

Business Deliverables

- Readmission Reduction Strategy
- Care Coordination Improvement Plan
- Financial Impact & ROI Assessment
- Executive Recommendations
- Board of Directors Presentation

2. INDUSTRY BACKGROUND BRIEF

Healthcare Industry Overview

The healthcare industry has undergone significant transformation over the past two decades.

Healthcare organizations are no longer evaluated solely by the volume of services they provide, but increasingly by the quality, efficiency, and outcomes of patient care.

Historically, hospitals were reimbursed primarily based on the number of admissions, procedures, and services delivered. Today's healthcare environment places growing emphasis on value-based care, patient outcomes, operational efficiency, and financial stewardship.

Healthcare organizations must now balance:

- High quality patient care
- Positive patient outcomes
- Financial sustainability
- Regulatory compliance
- Operational efficiency
- Patient satisfaction

One of the most important quality indicators used to evaluate healthcare organizations is the hospital readmission rate.

Why Hospital Readmissions Matter

Hospital readmissions are among the most closely monitored performance indicators within the healthcare industry because they often reflect the effectiveness of patient care beyond the hospital stay.

Readmission occurs when a patient returns to the hospital within a specified period after discharge.

The most commonly monitored quality metric is the:
30-Day Hospital Readmission Rate
Patients readmitted within 30 days following discharge.

Although some readmissions are unavoidable, many are considered preventable and may indicate opportunities for improvement.

Common contributors include:

- Inadequate discharge planning
- Poor care coordination
- Incomplete medication management
- Limited follow-up care
- Chronic disease management challenges
- Social determinants of health
- Limited access to community resources

Reducing avoidable readmissions improves both patient outcomes and organizational performance.

CMS Hospital Readmissions Reduction Program (HRRP)

The Centers for Medicare & Medicaid Services (CMS) established the Hospital Readmissions Reduction Program (HRRP) to encourage hospitals to improve quality of care while reducing preventable readmissions.

Hospitals experiencing excessive readmission rates may face:

- Reduced Medicare reimbursement
- Financial penalties
- Lower quality rankings
- Increased regulatory oversight
- Greater public scrutiny

As a result, hospital readmission reduction has become both a clinical priority and a financial priority for healthcare organizations nationwide.

Current Industry Challenges

Healthcare organizations face numerous challenges that directly influence patient outcomes and operational performance.

Aging Population

Older patients frequently experience:

- Multiple chronic conditions
- Increased medication complexity
- Longer recovery periods
- Greater risk of readmission
- Increased utilization of healthcare services

Care Coordination

Patients frequently receive care from multiple providers and healthcare settings.

Breakdowns in communication between providers may contribute to:

- Medication errors
- Missed follow-up appointments
- Duplicate testing
- Poor discharge transitions
- Increased readmission risk

Workforce Challenges

Healthcare organizations continue facing shortages of:

- Physicians
- Registered nurses
- Care coordinators
- Primary care providers
- Case managers

Limited staffing can directly affect patient care, discharge planning, and follow-up services.

Rising Healthcare Costs

Healthcare expenditures continue to increase while reimbursement rates remain constrained.

Executive leadership must continually balance:

- Patient outcomes
- Operational efficiency
- Financial performance
- Resource allocation

Organizations that successfully reduce preventable readmissions often improve both quality of care and financial performance.

Health Equity

Healthcare organizations increasingly recognize that patient outcomes are influenced by more than clinical care alone.

Differences in outcomes frequently exist across:

- Geographic regions
- Insurance coverage
- Socioeconomic populations
- Access to transportation
- Access to primary care
- Demographic populations

Understanding these factors is essential when designing effective interventions.

Data Driven Healthcare

Modern healthcare organizations collect enormous amounts of information through:

- Electronic Health Records (EHRs)
- Patient demographics
- Clinical diagnoses
- Care coordination programs
- Financial systems
- Quality reporting initiatives
- Population health management systems

Although these data provide tremendous opportunities, many organizations struggle to integrate clinical, operational, and financial information into meaningful business intelligence.

This consulting engagement provides an opportunity to combine patient level clinical information, financial impact data, and care coordination data to develop a comprehensive understanding of hospital readmissions and identify opportunities for improving patient outcomes and organizational performance.

Key Terminology

Readmission

A patient returning to the hospital within a specified period following discharge.

30-Day Readmission Rate

The percentage of patients readmitted within 30 days after discharge.

Value Based Care

A healthcare reimbursement model emphasizing patient outcomes, quality, and efficiency rather than service volume.

Care Coordination

The organization and management of patient care activities across multiple providers and healthcare settings to improve patient outcomes.

Chronic Disease Management

The ongoing treatment and monitoring of long-term health conditions to reduce complications and prevent unnecessary hospitalizations.

Risk Stratification

The process of identifying patients at varying levels of clinical risk to prioritize interventions and allocate healthcare resources effectively.

Return on Investment (ROI)

The estimated financial value generated through investments in quality improvement, care coordination, and operational initiatives.

Why This Matters to Your Consulting Engagement

Throughout this consulting engagement, your team will integrate three different datasets representing clinical outcomes, financial performance, and care coordination activities.

Executive Leadership expects your team to determine how these business areas interact and identify opportunities where improvements in patient care, discharge planning, financial management, and care coordination can collectively reduce hospital readmissions and improve organizational performance.

Your recommendations should demonstrate not only what is happening within National Hospital Network, but why it is happening and how leadership can respond using data-driven decision-making.

3. BUSINESS PROBLEM OVERVIEW

National Hospital Network has experienced a steady increase in 30-day hospital readmissions over the past four years. While some readmissions are expected within any healthcare system, the current trend has become a significant clinical, operational, and financial concern for Executive Leadership.

Although NHN has invested in discharge planning, care coordination initiatives, quality improvement programs, and patient follow-up services, readmission rates have continued to rise across multiple patient populations and service lines.

Executive Leadership believes the organization possesses sufficient clinical, financial, and operational data to identify the root causes of these readmissions, but those data have never been analyzed together as a single organizational problem.

This consulting engagement has been commissioned to determine why readmissions continue to increase, identify opportunities for improvement, and recommend data driven strategies that improve patient outcomes while reducing avoidable healthcare costs.

Historical Performance

Year Readmission Rate

2022 14.2%

2023 15.8%

2024 16.9%

2025 18.4%

Although the annual increases appear incremental, the cumulative impact has significantly affected organizational performance, resource utilization, and financial sustainability. Executive Leadership believes this trend cannot continue without affecting the hospital network's long-term quality objectives and financial performance.

Estimated Financial Impact

Executive Leadership estimates annual readmission related costs exceed: \$42 Million

Potential sources include:

- Additional inpatient stays
- Increased resource utilization
- Administrative costs
- CMS reimbursement penalties
- Care coordination costs
- Increased staffing requirements
- Reduced operational efficiency
- Lost opportunities to improve patient throughput

Leadership believes reducing even a portion of avoidable readmissions would create substantial clinical and financial value while improving patient care.

Executive Concerns

Executive Leadership believes rising readmission rates are influenced by multiple interconnected clinical and operational challenges rather than a single issue.

Rather than assuming one root cause, leadership has identified several working theories that require investigation.

Theory 1

Discharge Planning May Not Adequately Prepare Patients for Recovery

Leadership believes inconsistencies in discharge planning may contribute to avoidable readmissions.

Potential concerns include:

- Incomplete discharge instructions
- Limited patient education
- Poor medication reconciliation
- Inadequate transition planning

Your consulting team should determine whether discharge related variables influence readmission risk.

Theory 2

Care Coordination Programs May Not Effectively Target High-Risk Patients

NHN has invested heavily in care coordination services.

However, Executive Leadership does not know:

- Which patients receive care coordination
- Whether current interventions reduce readmissions
- Which services provide the greatest benefit

The Care Coordination Dataset has been provided to evaluate these questions.

Theory 3

Chronic Disease Management May Require Improvement

Patients with multiple chronic conditions frequently experience higher rates of hospitalization and readmission.

Leadership wants to determine whether:

- Chronic conditions increase readmission risk
- Certain diagnoses require additional intervention
- Specific patient populations should receive enhanced follow-up care

Theory 4

Financial Resources May Not Be Directed Toward the Highest-Impact Interventions

Executive Leadership has invested considerable resources in quality improvement initiatives.

However, they do not know:

- Which interventions provide measurable financial value
- Which programs reduce readmissions most effectively
- Where future investments should be prioritized

The Financial Impact Dataset has been provided to support cost benefit and return on investment analyses.

Theory 5

Patient Follow-Up and Community Support May Influence Readmission Risk

Leadership believes post discharge support services may significantly influence patient outcomes.

Areas of concern include:

- Follow-up appointment completion
- Medication review
- Home health referrals
- Transportation assistance
- Post discharge communication

Your consulting team should evaluate which services appear most effective in reducing avoidable readmissions.

Theory 6

Organizational Decision-Making Lacks Integrated Clinical, Financial, and Operational Analytics

Clinical outcomes, financial performance, and care coordination activities are currently analyzed independently.

Executive Leadership believes integrating these perspectives will provide a more complete understanding of readmissions and improve strategic decision making.

One objective of this consulting engagement is to combine patient level clinical information, financial impact data, and care coordination activities into a unified analytical framework capable of supporting executive decisions.

Consulting Challenge

Your consulting team has been asked to determine:

- Why are hospital readmissions increasing?
- Which patients are at greatest risk of readmission?
- Which care coordination interventions are most effective?
- Which patient populations should receive additional resources?
- Which investments will produce the greatest improvement in patient outcomes?
- How can National Hospital Network reduce readmissions while improving both clinical quality and financial performance?

Your recommendations should be supported through:

- Clinical data analysis
- Patient risk stratification
- Predictive machine learning models
- Financial impact assessment
- Care coordination analysis
- Operational analysis
- Executive dashboards
- Return on Investment (ROI) analysis

4. ORGANIZATIONAL BACKGROUND

National Hospital Network

Company Overview

National Hospital Network (NHN) is one of the largest integrated healthcare systems in the United States, providing comprehensive inpatient, outpatient, emergency, and specialty healthcare services to millions of patients each year.

NHN operates a diverse network of hospitals, outpatient clinics, specialty care centers, and community health facilities that serve urban, suburban, and rural populations across multiple states.

Over the past decade, NHN has invested heavily in electronic health records, population health management, care coordination initiatives, quality improvement programs, predictive analytics, and data driven clinical decision making.

Today, the organization is focused on improving patient outcomes while simultaneously enhancing operational efficiency and long-term financial sustainability.

Company Facts

- 42 hospitals
- More than 300 outpatient facilities
- Operations across 11 states
- 2.3 million patients annually
- 48,000 employees
- Comprehensive inpatient and outpatient care
- Integrated care coordination programs
- Enterprise Electronic Health Record (EHR) system

Mission

Improve patient outcomes through evidence-based care, innovation, collaboration, and data driven decision making while delivering safe, high-quality healthcare to every patient.

Vision

To become a national leader in patient centered healthcare by combining clinical excellence, operational efficiency, advanced analytics, and innovative care delivery to improve the health of the communities we serve.

Strategic Priorities

Executive Leadership has identified the following strategic priorities for the next three years.

Priority 1

Reduce avoidable hospital readmissions by improving discharge planning, patient education, and care coordination.

Priority 2

Improve patient outcomes through earlier intervention, predictive analytics, and evidence based clinical decision making.

Priority 3

Enhance operational efficiency by improving resource utilization, reducing unnecessary costs, and optimizing patient flow throughout the healthcare system.

Priority 4

Strengthen population health initiatives through proactive patient management, improved chronic disease management, and expanded community-based healthcare services.

Why This Consulting Engagement Matters

Executive Leadership believes National Hospital Network possesses extensive clinical, financial, and operational information capable of improving patient outcomes and organizational performance.

However, these data sources currently exist within separate clinical and administrative systems and have rarely been analyzed together as part of a comprehensive strategic review.

This consulting engagement provides an opportunity to integrate:

- Clinical information
- Patient outcomes
- Care coordination activities
- Financial performance
- Operational efficiency

into a unified analytical framework that supports executive decision making.

The Board of Directors views this engagement as an opportunity to identify the clinical and operational initiatives that will have the greatest impact on reducing avoidable readmissions while improving quality of care and financial sustainability.

Organizational Success Will Be Measured By

At the conclusion of this consulting engagement, Executive Leadership expects recommendations that help National Hospital Network:

- Reduce avoidable hospital readmissions.
- Improve patient outcomes.
- Strengthen care coordination.
- Improve discharge planning.
- Better allocate clinical resources.
- Improve operational efficiency.
- Reduce avoidable healthcare costs.
- Support long term population health initiatives.

5. DATASET DESCRIPTION

Your consulting team has been provided with three integrated datasets representing the clinical, financial, and operational dimensions of patient care across National Hospital Network.

These datasets have been assembled to simulate the information available to a professional healthcare consulting team engaged to evaluate patient outcomes, operational effectiveness, and organizational performance.

Rather than answering a single clinical question, these datasets provide complementary perspectives on patient care, financial performance, and care coordination.

Throughout this consulting engagement, your team is expected to integrate information from all three data sets to develop a comprehensive understanding of hospital readmissions and identify opportunities for improving patient outcomes while reducing avoidable healthcare costs.

Dataset 1: Patient Readmission Dataset

Purpose

The Patient Readmission Dataset serves as the primary clinical dataset for this consulting engagement.

This dataset should be used to identify patients at high risk of readmission, evaluate clinical factors influencing patient outcomes, and support predictive modeling activities.

Your patient risk analysis and machine learning activities should primarily begin with this dataset.

Contains:

Patient Information

- Patient ID
- Age
- Gender
- Insurance Type

Clinical Information

- Primary Diagnosis
- Chronic Conditions
- Prior Admissions

Hospitalization Information

- Length of Stay
- Admission Type
- Discharge Disposition

Utilization Information

- Emergency Department Visits
- Follow-Up Status

Outcome Variable

- Readmitted Within 30 Days

Dataset 2: Financial Impact Dataset

Purpose

The Financial Impact Dataset provides information regarding the financial consequences of hospital readmissions.

Executive Leadership believes reducing readmissions should improve both patient outcomes and organizational financial performance.

This dataset should be used to estimate:

- Readmission costs
- CMS reimbursement penalties
- Total care costs
- Return on investment (ROI)

Contains:

Financial Information

- Readmission Cost
- Reimbursement Amount
- Penalty Cost
- Length of Stay Cost
- Follow-Up Program Cost
- Total Care Cost

Dataset 3: Care Coordination Dataset

Purpose

The Care Coordination Dataset provides operational information regarding post discharge interventions and patient follow-up activities.

Executive Leadership believes improvements in care coordination may represent one of the greatest opportunities for reducing avoidable readmissions.

This dataset should be used to evaluate:

- Care coordination effectiveness
- Follow-up completion
- Medication review
- Home health referrals
- Transportation assistance
- Post discharge communication

Contains:

Care Coordination Information

- Care Coordinator Assigned
- Follow-Up Completed
- Follow-Up Days After Discharge
- Medication Review Completed
- Home Health Referral
- Transportation Assistance Provided
- Post Discharge Calls

Data Relationships

Although each dataset examines a different aspect of patient care, Executive Leadership expects your consulting team to integrate findings across all three datasets.

The datasets should be viewed as complementary components of a larger organizational challenge rather than independent sources of information.

Patient Readmission Dataset

Provides clinical information and patient outcomes.

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Financial Impact Dataset

Explains the financial consequences associated with patient outcomes.



Care Coordination Dataset

Provides operational context for post discharge interventions and patient support.

Together, these datasets allow your consulting team to evaluate hospital readmissions from clinical, financial, and operational perspectives while identifying opportunities to improve patient outcomes and organizational performance.

Data Quality Notes

Executive Leadership believes several data quality challenges exist within the available information.

These include:

- Missing values within clinical records
- Incomplete follow-up documentation
- Financial reporting inconsistencies
- Missing care coordination information
- Variability in documentation across facilities

Part of your consulting engagement is evaluating these data quality issues, documenting their potential impact on analysis, and determining appropriate methods for handling incomplete or inconsistent information.

Consulting Recommendation

Executive Leadership expects your consulting team to determine:

- Which clinical factors contribute most significantly to hospital readmissions.
- Which patient populations require additional intervention.
- Which care coordination initiatives produce the greatest improvement in patient outcomes.
- Which investments provide the greatest financial return.
- How clinical, financial, and operational information can be integrated to support executive decision-making.

Your recommendations should demonstrate how National Hospital Network can improve patient outcomes through coordinated improvements in clinical care, financial management, discharge planning, and care coordination while creating measurable long term organizational value.

6. EXECUTIVE QUESTIONS

The Executive Leadership Team expects your consulting team to provide evidence-based answers to the following questions. Your recommendations should be supported through clinical analysis, predictive modeling, financial evaluation, care coordination analysis, and executive dashboard development.

Your consulting team is expected to integrate findings from all three datasets to develop comprehensive recommendations that improve patient outcomes while reducing avoidable hospital readmissions and strengthening National Hospital Network's long term operational and financial performance.

Question 1

Which patients are most likely to be readmitted?

Identify the demographic, clinical, behavioral, and operational characteristics that distinguish patients at highest risk for readmission.

Consider factors such as:

- Patient age
- Primary diagnosis
- Chronic conditions
- Prior hospital admissions
- Length of stay
- Insurance type
- Emergency department utilization
- Follow-up status

Support your conclusions with descriptive statistics, visualizations, and patient segmentation.

Question 2

What factors contribute most significantly to hospital readmission risk?

Determine which variables have the strongest influence on patient readmissions.

Examples include:

- Clinical diagnoses
- Chronic disease burden
- Prior admissions
- Length of stay
- Discharge disposition
- Follow-up care
- Care coordination
- Patient demographics

Support your conclusions using statistical analysis, feature importance, and business interpretation.

Question 3

Can we accurately identify high-risk patients before discharge?

Develop predictive machine learning models capable of identifying patients who are at greatest risk of readmission.

Compare multiple modeling approaches and determine:

- Which model performs best?
- Which variables are most predictive?
- What limitations exist?
- How should leadership use these predictions?

Executive Leadership expects a model that is both accurate and explainable for use within a clinical environment.

Question 4

Which patient populations should receive additional care coordination and intervention?

Using the Patient Readmission Dataset and the Care Coordination Dataset, determine:

- Which patient groups require additional follow-up?
- Which patients should receive care coordination services?
- Which populations benefit most from post discharge interventions?
- Which services should receive additional organizational investment?

Recommendations should balance clinical outcomes, operational feasibility, and financial impact.

Question 5

Which care coordination interventions appear most effective?

Using the Care Coordination Dataset, evaluate the effectiveness of current patient support programs.

Examples include:

- Care coordinators
- Follow-up calls
- Home health referrals
- Medication reviews
- Transportation assistance
- Post-discharge communication

Determine:

- Which interventions reduce readmission risk?
- Which interventions should be expanded?
- Which interventions should be redesigned?
- Which interventions provide the greatest clinical value?

Support your recommendations using quantitative evidence.

Question 6

What financial savings could be achieved by reducing avoidable hospital readmissions?

Using the Financial Impact Dataset, estimate the organizational value associated with reducing readmissions.

Your analysis should estimate:

- Reduced readmissions
- Avoided hospitalization costs
- Reduced CMS reimbursement penalties
- Reduced total care costs
- Improved resource utilization

- Expected Return on Investment (ROI)

Whenever possible, provide:

- Conservative estimate
- Expected estimate
- Optimistic estimate

Discuss implementation costs and expected payback period.

Question 7

What strategic initiatives should Executive Leadership prioritize during the next 12 months?

Based on your complete analysis, recommend a prioritized action plan.

Your recommendations should address:

- Patient risk identification
- Care coordination
- Discharge planning
- Medication management
- Follow-up care
- Resource allocation
- Financial management
- Executive reporting and dashboards

Recommendations should be realistic, measurable, financially justified, and aligned with National Hospital Network's strategic priorities.

Executive Consulting Challenge

Throughout this consulting engagement, remember that the Executive Leadership Team is not asking your consulting team to simply predict hospital readmissions.

They are asking a much broader executive question:

How can National Hospital Network use clinical analytics, predictive modeling, financial analysis, and care coordination intelligence to reduce avoidable hospital readmissions, improve patient outcomes, optimize operational performance, and create measurable long term organizational value?

Every analysis, visualization, predictive model, dashboard, financial estimate, and recommendation should contribute toward answering that question.

7. PROJECT SCOPE

Your consulting team has been retained to conduct a comprehensive evaluation of patient outcomes, hospital readmissions, care coordination effectiveness, financial performance, and operational efficiency across National Hospital Network.

The Executive Leadership Team expects your recommendations to be supported by rigorous clinical analysis, predictive modeling, financial evaluation, operational assessment, and executive level communication.

Throughout this consulting engagement, your team should integrate findings from all three data sets to develop a comprehensive understanding of the factors contributing to hospital readmissions and identify opportunities to improve patient outcomes while reducing avoidable healthcare costs.

Exploratory Analysis

Required

Your consulting team should begin by developing a thorough understanding of the available data and identifying opportunities for further investigation.

Required activities include:

- Data quality assessment
- Missing value analysis
- Duplicate record identification
- Outlier detection
- Patient segmentation
- Readmission trend analysis
- Clinical diagnosis analysis
- Care coordination analysis
- Financial impact exploration
- Operational trend analysis

You should summarize:

- Major trends
- Unexpected findings
- Data limitations
- Initial business hypotheses

Predictive Modeling

Your consulting team should develop predictive models capable of identifying patients at greatest risk of hospital readmission.

Baseline Model

Examples include:

- Logistic Regression
- Decision Tree

The baseline model should establish a benchmark against which more advanced models can be evaluated.

Advanced Model

Examples include:

- Random Forest
- Gradient Boosting
- XGBoost

You are encouraged to compare multiple modeling approaches and justify the selection of the final model.

Model Evaluation

Your consulting team should evaluate each predictive model using appropriate performance metrics.

Examples include:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Confusion Matrix

Executive Leadership expects your team to explain:

- Why the selected model performs well
- Which variables have the greatest influence
- Whether the model is appropriate for operational use
- What limitations should be considered before implementation

Potential Modeling Goals

Depending on your analytical approach, predictive models may focus on:

- Hospital readmission prediction
- Patient risk stratification
- Length of stay prediction
- Care coordination effectiveness
- Resource utilization prediction

Your team should clearly explain why the selected modeling objective best supports National Hospital Network's business objectives.

Clinical Analysis

Your consulting team should evaluate the clinical factors contributing to patient readmissions.

Required analyses include:

- Patient demographics
- Primary diagnoses
- Chronic disease burden

- Length of stay
- Prior admissions
- Emergency department utilization
- Readmission patterns

Identify:

- High risk patient populations
- Clinical risk factors
- Opportunities for earlier intervention

Financial Analysis

Using the Financial Impact Dataset, evaluate the financial consequences associated with hospital readmissions.

Required analyses include:

- Readmission costs
- CMS reimbursement penalties
- Total care costs
- Resource utilization
- Intervention costs
- Return on Investment (ROI)

You should estimate:

- Cost savings
- Financial impact
- Return on Investment
- Expected organizational value

Care Coordination Analysis

Using the Care Coordination Dataset, evaluate the effectiveness of current patient support programs.

Required analyses include:

- Care coordinator assignment
- Follow-up completion
- Medication review
- Home health referrals
- Transportation assistance
- Post discharge communication

Determine:

- Which interventions appear most effective
- Which interventions should be expanded
- Which interventions require improvement

Operational Analysis

Your consulting team should evaluate how operational practices influence patient outcomes.

Examples include:

- Resource utilization
- Patient flow
- Care coordination efficiency
- Follow-up performance
- Clinical workflow effectiveness

Explain how operational improvements could reduce avoidable readmissions.

Executive Trend Analysis

Your consulting team should integrate findings from all three datasets to evaluate organizational performance.

Examples include:

- Readmission trends
- Patient population trends
- Care coordination trends
- Financial trends
- Operational performance trends

Explain how these trends influence executive decision making and long-term organizational strategy.

Dashboard Development

Using:

- Tableau
- Power BI

Develop an executive dashboard designed to support strategic decision making.

Required Dashboard Sections

Executive KPI Summary

Include:

- Readmission Rate
- Patient Risk Score
- Total Care Cost
- CMS Penalties
- Return on Investment (ROI)

Clinical Analytics

Include:

- Patient Risk Profiles
- Readmission Trends
- Diagnosis Categories
- Chronic Disease Burden

Financial Analytics

Include:

- Readmission Costs
- Financial Impact
- CMS Penalties
- ROI

Care Coordination Analytics

Include:

- Follow-up Completion
- Care Coordinator Assignment
- Medication Reviews
- Home Health Referrals

Executive Recommendation Center

Summarize:

- Highest priority recommendations
- Expected clinical improvements
- Expected financial value
- Estimated ROI
- Suggested implementation priorities

Consulting Expectations

Throughout this engagement, remember that Executive Leadership is not asking for the most technically sophisticated predictive model.

Instead, they expect your consulting team to deliver:

- Data driven clinical insights
- Evidence based recommendations
- Financial justification
- Executive ready dashboards
- Actionable operational strategies

Every analytical decision should support better executive decision making and demonstrate how data science can improve patient outcomes, strengthen care coordination, reduce avoidable healthcare costs, and enhance the long-term performance of National Hospital Network.

8. DELIVERABLES

Throughout this consulting engagement, your team will produce a series of technical and business deliverables that document your analytical process and support your final recommendations to Executive Leadership.

Each deliverable builds upon the previous one and should demonstrate how your team transformed clinical, financial, and operational data into actionable business intelligence that improves patient outcomes and organizational performance.

Deliverable 1

Project Charter

Develop a consulting project charter that clearly defines your team's approach to the engagement. Include:

- Business Problem Statement
- Project Objectives
- Stakeholders
- Scope of Work
- Initial Business Hypotheses
- Success Metrics
- Anticipated Risks
- Team Roles and Responsibilities

The Project Charter serves as the roadmap for the remainder of the consulting engagement.

Deliverable 2

Exploratory Data Analysis (EDA) Report

Conduct a comprehensive exploratory analysis across all three datasets.

Your report should summarize:

Data Quality Assessment

- Missing values
- Duplicate records
- Outliers
- Data inconsistencies

Clinical Analysis

- Patient demographics
- Diagnosis patterns
- Chronic disease burden
- Readmission trends

Financial Analysis

- Readmission costs
- CMS penalties
- Total care costs
- Resource utilization

Care Coordination Analysis

- Care coordinator assignments
- Follow-up completion
- Medication review
- Home health referrals
- Transportation assistance

Trend Analysis

- Patient outcome trends
- Clinical trends
- Operational trends

- Financial trends

Minimum Requirements

- Five professional visualizations
- Key findings
- Areas requiring further investigation
- Initial business hypotheses supported by evidence

Deliverable 3

Model Evaluation Report

Develop and compare predictive models designed to identify patients at greatest risk of hospital readmission.

Include:

Model Selection

- Baseline model
- Advanced model
- Rationale for model selection

Performance Metrics

Examples:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Confusion Matrix

Model Interpretation

Include:

- Feature importance
- Variable interpretation
- Model limitations
- Business implications

Executive Leadership expects recommendations supported by both statistical performance and practical clinical value.

Deliverable 4

Clinical & Care Coordination Analysis Report

Using the Patient Readmission Dataset and the Care Coordination Dataset, evaluate patient outcomes and the effectiveness of current intervention strategies.

Include:

Patient Risk Analysis

- High risk patient populations

- Readmission trends
- Diagnosis categories
- Chronic disease burden

Care Coordination Analysis

- Care coordinator effectiveness
- Follow-up completion
- Medication review
- Home health referrals
- Transportation assistance
- Post discharge communication

Strategic Recommendations

Identify:

- Which interventions should be expanded
- Which interventions should be redesigned
- Which patient populations should receive additional resources
- Which operational improvements should receive priority

Support all recommendations using quantitative evidence.

Deliverable 5

Financial Impact & ROI Analysis Report

Quantify the financial value associated with your recommendations.

Your report should estimate:

Financial Benefits

- Reduced readmission costs
- Reduced CMS penalties
- Reduced total care costs
- Improved resource utilization
- Increased operational efficiency

Investment Analysis

Estimate:

- Implementation costs
- Expected financial benefits
- Return on Investment (ROI)

Whenever possible, provide:

- Conservative estimate
- Expected estimate
- Optimistic estimate

Deliverable 6

Executive Dashboard

Develop an interactive executive dashboard using Tableau or Power BI.

Your dashboard should support executive decision-making and include the following sections.

Executive KPI Summary

Include:

- Readmission Rate
- Patient Risk Score
- Total Care Cost
- CMS Penalties
- Return on Investment (ROI)

Clinical Analytics

- Patient Risk Profiles
- Diagnosis Categories
- Chronic Disease Burden
- Readmission Trends

Financial Analytics

- Readmission Costs
- Financial Impact
- CMS Penalties
- Resource Utilization

Care Coordination Analytics

- Follow-up Completion
- Care Coordinator Assignment
- Medication Reviews
- Home Health Referrals
- Transportation Assistance

Executive Recommendation Center

Summarize:

- Highest priority recommendations
- Expected clinical improvements
- Expected financial value
- Estimated ROI
- Suggested implementation priorities

Deliverable 7

Executive Board Presentation

Prepare and deliver an executive presentation summarizing your consulting engagement.

Your presentation should communicate technical findings in language appropriate for Executive Leadership and the Board of Directors.

Your presentation should include:

- Business Problem
- Industry Overview
- Data Overview
- Key Analytical Findings
- Predictive Modeling Results
- Clinical Analysis
- Financial Analysis
- Care Coordination Analysis
- Executive Dashboard Demonstration
- Strategic Recommendations
- Implementation Roadmap
- Expected Clinical Impact
- Expected Financial Value
- Return on Investment (ROI)

The purpose of the presentation is not to demonstrate technical expertise alone, but to convince Executive Leadership that your recommendations are practical, financially justified, clinically meaningful, and capable of improving patient outcomes while reducing avoidable healthcare costs and strengthening National Hospital Network's long term organizational performance.

9. CONSTRAINTS

The Executive Leadership Team has established the following constraints for this consulting engagement.

Your recommendations should be realistic, clinically appropriate, financially responsible, operationally feasible, and capable of being implemented within National Hospital Network's existing healthcare environment.

While innovation is encouraged, all recommendations must recognize the clinical, regulatory, financial, and operational realities facing a large integrated healthcare system.

Budget

Executive Leadership expects all recommendations to be financially feasible and supported by measurable business value.

Priority should be given to initiatives that:

- Reduce avoidable readmissions
- Improve patient outcomes
- Improve care coordination
- Increase operational efficiency
- Reduce unnecessary healthcare costs

- Generate measurable Return on Investment (ROI)

Recommendations requiring significant capital investment should include a clear financial justification and expected ROI.

Staffing

National Hospital Network continues to experience workforce shortages across multiple clinical and administrative areas.

Recommendations should recognize limitations in:

- Nursing staff
- Physicians
- Care coordinators
- Case managers
- Social workers
- Primary care providers

Whenever possible, recommendations should improve efficiency by leveraging existing clinical teams rather than relying exclusively on additional staffing.

Regulatory Compliance

All recommendations must comply with applicable healthcare regulations and patient privacy requirements.

Examples include:

- HIPAA
- CMS Hospital Readmissions Reduction Program (HRRP)
- Medicare Conditions of Participation
- Patient privacy regulations
- Healthcare quality reporting requirements

Predictive models and patient risk stratification approaches should be transparent, explainable, and appropriate for use within a regulated healthcare environment.

Technology

Executive Leadership prefers recommendations that leverage existing technology investments whenever practical.

Solutions should integrate with existing:

- Electronic Health Record (EHR) systems
- Care coordination platforms
- Population health management systems
- Clinical reporting systems
- Business intelligence platforms

Recommendations requiring significant technology replacement should clearly justify the expected organizational value.

Patient Experience

Executive Leadership believes patient centered care remains one of the organization's highest priorities.

Recommendations should improve patient outcomes without negatively impacting the patient experience.

Examples include:

- Improving discharge planning
- Increasing follow-up compliance
- Supporting patient education
- Improving communication
- Increasing continuity of care

Recommendations should strengthen patient trust while improving organizational performance.

Operational Disruption

Recommendations should minimize disruption to daily clinical operations.

Your consulting team should consider the potential impact of proposed initiatives on:

- Patient care
- Hospital operations
- Clinical workflows
- Care coordination teams
- Provider workload

Whenever possible, recommend phased implementation strategies that allow improvements to be introduced with minimal disruption.

Data Limitations

Executive Leadership recognizes that the available data sets contain limitations.

Your consulting team should document:

- Missing values
- Incomplete patient records
- Missing care coordination documentation
- Financial reporting inconsistencies
- Variability across facilities

Recommendations should acknowledge these limitations and explain how they may influence analytical findings and business decisions.

Implementation Timeline

Executive Leadership expects recommended initiatives to be achievable within 12 months.

For each recommendation, estimate:

- Implementation timeline
- Required organizational resources
- Potential implementation risks
- Expected clinical impact
- Expected financial impact

Recommendations requiring longer implementation periods should include a phased implementation strategy.

Organizational Resources

Leadership recognizes that financial, clinical, operational, and human resources are limited.

Your recommendations should prioritize initiatives that:

- Produce the greatest improvement in patient outcomes
- Maximize Return on Investment
- Improve care coordination
- Can be supported by existing clinical teams
- Deliver measurable improvements within the first year

Consulting Expectations

Throughout this engagement, remember that Executive Leadership is not looking for the most technically sophisticated predictive model.

Instead, they expect recommendations that are:

- Data driven
- Clinically appropriate
- Financially justified
- Operationally feasible
- Patient centered
- Realistic to implement
- Aligned with National Hospital Network's strategic priorities

Your team's ability to balance clinical quality, financial stewardship, and operational efficiency will be one of the primary factors used to evaluate the success of this consulting engagement.

10. SUCCESS METRICS

The Executive Leadership Team will evaluate the success of this consulting engagement based on the quality of your analysis, the practicality of your recommendations, and your ability to translate complex analytical findings into meaningful clinical and business value.

Your recommendations should demonstrate how National Hospital Network can improve patient outcomes while strengthening operational efficiency and long-term financial sustainability.

Executive Leadership expects your consulting team to provide recommendations that are measurable, actionable, financially justified, clinically appropriate, and aligned with the organization's strategic priorities.

Clinical Outcomes

Your recommendations should demonstrate how National Hospital Network can improve patient care and reduce avoidable hospital readmissions.

Examples include:

- Reduced hospital readmission rate
- Improved patient outcomes
- Improved chronic disease management
- Improved patient follow-up
- Improved medication management
- Improved discharge planning
- Improved continuity of care

Whenever possible, estimate expected improvements in patient outcomes.

Financial Outcomes

Executive Leadership expects recommendations that produce measurable financial value.

Examples include:

- Reduced CMS reimbursement penalties
- Reduced readmission costs
- Reduced total care costs
- Improved resource utilization
- Increased operational efficiency
- Positive Return on Investment (ROI)

Whenever possible, estimate:

- Conservative financial impact
- Expected financial impact
- Optimistic financial impact

Operational Outcomes

Your consulting team should identify operational improvements that enhance patient care while improving organizational efficiency.

Examples include:

- Improved care coordination
- Better intervention effectiveness
- Improved resource utilization
- Improved patient flow
- Improved discharge planning
- Better allocation of clinical resources

- Improved executive decision-making through analytics

Analytics Outcomes

Executive Leadership expects recommendations supported by rigorous data science methods.

Successful consulting engagements should include:

- Accurate predictive models
- Clearly interpreted model results
- Executive ready dashboards
- Effective patient risk stratification
- Meaningful care coordination analysis
- Financial impact assessment
- Actionable business recommendations

Technical sophistication is important, but leadership places greater value on analyses that are understandable, explainable, and capable of supporting executive decision making.

Executive Communication Outcomes

As a consulting team, your recommendations should demonstrate your ability to communicate complex analytical findings to non-technical audiences.

Executive Leadership will evaluate your ability to:

- Translate technical findings into business language
- Prioritize recommendations based on patient and organizational value
- Defend recommendations using evidence from the data
- Balance analytical rigor with practical implementation
- Communicate clearly and confidently with executive stakeholders

FINAL PRESENTATION PROMPT

You are presenting the results of your consulting engagement to the National Hospital Network Board of Directors.

The Board has limited time and is not interested in technical implementation details or algorithm selection.

Instead, they expect your consulting team to answer one fundamental business question:

How can National Hospital Network use clinical analytics, predictive modeling, financial analysis, and care coordination intelligence to reduce avoidable hospital readmissions, improve patient outcomes, optimize operational performance, and create measurable long term organizational value?

Your presentation should demonstrate how your team transformed clinical, financial, and operational data into actionable business intelligence by integrating:

- Clinical analysis

- Patient risk stratification
- Predictive machine learning models
- Financial impact analysis
- Care coordination effectiveness
- Executive dashboards
- Strategic recommendations
- Return on Investment (ROI)

Every chart, visualization, predictive model, dashboard, and recommendation should support that executive question.

Executive Presentation Expectations

Your Board presentation should demonstrate that your consulting team can:

- Translate complex analytical findings into executive level recommendations.
- Integrate multiple datasets into a single clinical and business narrative.
- Balance analytical rigor with practical healthcare decision making.
- Quantify the clinical and financial value of your recommendations.
- Clearly communicate complex information to non-technical stakeholders.
- Defend your recommendations using evidence from the data.

The Executive Leadership Team will evaluate your work based on the quality of your analysis, the practicality of your recommendations, and your ability to communicate how data science can improve patient outcomes, reduce avoidable readmissions, strengthen care coordination, and improve the long term performance of National Hospital Network.